

ASE LAB 9

Perform Static Code Analysis of a Java Program Containing Duplicate Code Statements Using SonarQube to Identity Code of Duplication and Maintainability Issues.

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Objective

To analyze a Java program using SonarQube and identify repeated or duplicate code using Copy-Paste Detection (CPD).

Step 1: Open Docker Desktop

Firstly, open Docker Desktop.

Run the SonarQube container using Command Prompt:

```
docker run -d --name sonarqube -e SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000 sonarqube:latest
```

Wait for the command to finish. Then check the running container:

```
docker ps
```

If the SonarQube container is shown in the output, the container is running successfully.

```
C:\Users\vgami>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
8944316e358d   sonarqube:latest  "/opt/sonarqube/dock..."  44 minutes ago  Up 10 seconds  0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp  sonarqube
C:\Users\vgami>
```

Step 2: Open SonarQube using localhost

Open a browser and enter:

```
http://localhost:9000
```

Login using:

Username: admin

Password: admin

Step 3: Create the SonarQube Project

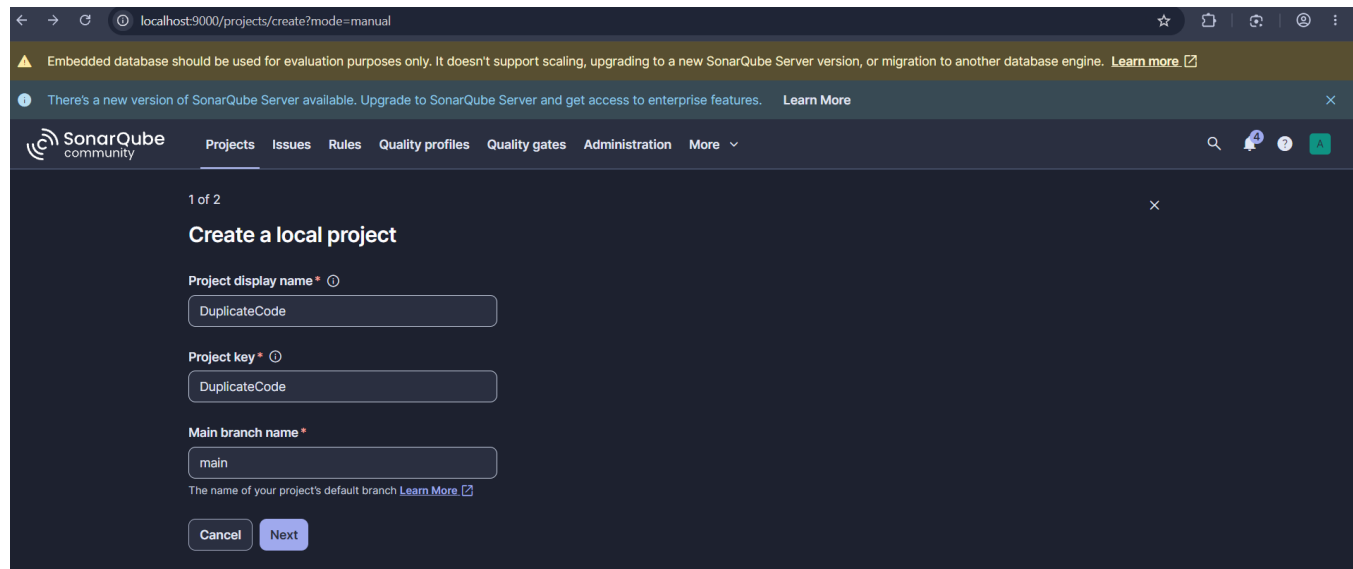
In SonarQube, create a new project.

Enter:

Project Name: DuplicateCode

Project Key: DuplicateCode

Keep the project public if that is the setting used in your class setup.



The screenshot shows the SonarQube web interface in a browser. The address bar shows 'localhost:9000/projects/create?mode=manual'. There are two warning banners at the top: one about the embedded database and another about a new server version. The navigation bar includes 'Projects', 'Issues', 'Rules', 'Quality profiles', 'Quality gates', 'Administration', and 'More'. The main content area is titled '1 of 2 Create a local project'. It contains three input fields: 'Project display name' with the value 'DuplicateCode', 'Project key' with the value 'DuplicateCode', and 'Main branch name' with the value 'main'. Below these fields is a note: 'The name of your project's default branch Learn More'. At the bottom are 'Cancel' and 'Next' buttons.

Step 4: Create the Java Project Folder

Create/open the Lab 9 folder:

```
D:\3-1\ASE\ASE LAB 9
```

The folder structure should be:

```
ASE LAB 9
├── sonar-project.properties
└── src
    └── Main.java
```

Create/open sonar-project.properties and add:

```
sonar.projectKey=DuplicateCode
sonar.projectName=DuplicateCode
sonar.sources=src
sonar.host.url=http://localhost:9000
```

Open:

```
D:\3-1\ASE\ASE LAB 9\src\Main.java
```

Use the following code. The same large block is intentionally repeated in two methods so SonarQube can detect duplicate code:

```
public class Main {

    public void calculateSalary() {

        int basic = 30000;
        int hra = 5000;
        int da = 3000;
        int ta = 2000;
        int medical = 1500;

        int total = basic + hra + da + ta + medical;

        System.out.println("Basic: " + basic);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("TA: " + ta);
        System.out.println("Medical: " + medical);
        System.out.println("Total: " + total);
        System.out.println("Salary calculated");
        System.out.println("Employee eligible");
        System.out.println("Salary processed");
    }

    public void calculateBonus() {

        int basic = 30000;
        int hra = 5000;
        int da = 3000;
        int ta = 2000;
        int medical = 1500;

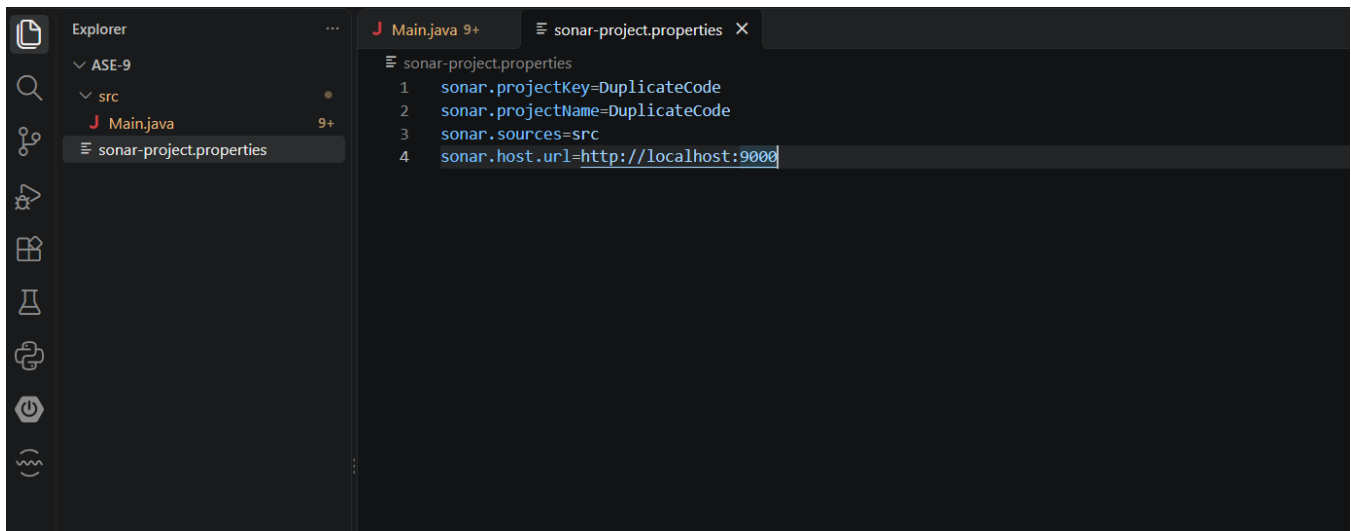
        int total = basic + hra + da + ta + medical;

        System.out.println("Basic: " + basic);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("TA: " + ta);
        System.out.println("Medical: " + medical);
        System.out.println("Total: " + total);
        System.out.println("Salary calculated");
        System.out.println("Employee eligible");
        System.out.println("Salary processed");
    }

    public static void main(String[] args) {
        Main obj = new Main();

        obj.calculateSalary();
        obj.calculateBonus();
    }
}
```

The code in calculateSalary() and calculateBonus() contains the same repeated block. This is the duplicate code that SonarQube will analyze.



Step 5: Install / Check SonarScanner

SonarScanner is used to send the Java project to SonarQube for analysis.

Check whether SonarScanner is installed:

```
sonar-scanner --version
```

You should see the SonarScanner version, for example:

```
SonarScanner CLI 8.1.0.6389
```

```
C:\Users\vgami>sonar-scanner --version
15:34:29.784 INFO Scanner configuration file: C:\Users\vgami\Downloads\sonar-scanner-cli-8.1.0.6389-windows-x64\
\conf\sonar-scanner.properties
15:34:29.786 INFO Project root configuration file: NONE
15:34:29.812 INFO SonarScanner CLI 8.1.0.6389
15:34:29.824 INFO Windows 11 10.0 amd64

C:\Users\vgami>
```

Step 6: Generate SonarQube Authentication Token

Go to SonarQube and login.

Then:

1. Open your profile from the top-right corner.
2. Open Security.
3. Enter a name for the token.
4. Select User Token if that option is shown.
5. Generate the token.
6. Copy the token and keep it safe. Do not share it.

Open Command Prompt and set the token for the current CMD window:

```
set SONAR_TOKEN=YOUR_TOKEN
```

Replace YOUR_TOKEN with the token you generated.

Step 7: Go to the Project Directory

Open Command Prompt and move to the Lab 9 project folder:

```
cd /d "D:\3-1\ASE\ASE LAB 9"
```

Check the files:

```
dir
```

You should see sonar-project.properties and the src folder.

Step 8: Run SonarScanner

Run:

```
sonar-scanner
```

Wait until the analysis is completed. The scanner should finish with:

```
EXECUTION SUCCESS
```

If you get an old .scannerwork permission error, run:

```
rmdir /s /q ".scannerwork"
```

Then run again:

```
sonar-scanner
```

If you get HTTP 401 Unauthorized, set SONAR_TOKEN again in the same CMD window and run sonar-scanner again.

```
C:\Users\vgami\OneDrive\Desktop\3rd Year\ODD\ASE\ASE-9>sonar-scanner
15:37:01.279 INFO Scanner configuration file: C:\Users\vgami\Downloads\sonar-scanner-cli-8.1.0.6389-windows-x64\sonar-scanner-8.1.0.6389-windows-x64\bin\..
\conf\sonar-scanner.properties
15:37:01.288 INFO Project root configuration file: C:\Users\vgami\OneDrive\Desktop\3rd Year\ODD\ASE\ASE-9\sonar-project.properties
15:37:01.312 INFO SonarScanner CLI 8.1.0.6389
15:37:01.325 INFO Windows 11 10.0 amd64
15:37:06.187 INFO Communicating with SonarQube Community Build 26.9.0.129388
15:37:06.189 INFO JRE provisioning: os[windows], arch[amd64]
15:37:07.114 INFO Starting SonarScanner Engine...
15:37:07.114 INFO Java 21.0.11 Eclipse Adoptium (64-bit)
15:37:12.902 INFO Load global settings
15:37:13.135 INFO Load global settings (done) | time=235ms
15:37:13.150 INFO Server id: 147B411E-AaELV_oVCzG6DkaFzVml
15:37:13.167 INFO Loading required plugins
15:37:13.168 INFO Load plugins index
15:37:13.217 INFO Load plugins index (done) | time=48ms
15:37:13.217 INFO Load/download plugins
15:37:13.271 INFO Load/download plugins (done) | time=54ms
15:37:13.933 INFO Process project properties
15:37:13.968 INFO Process project properties (done) | time=35ms
15:37:13.997 INFO Project key: DuplicateCode
15:37:13.998 INFO Base dir: C:\Users\vgami\OneDrive\Desktop\3rd Year\ODD\ASE\ASE-9
15:37:13.999 INFO Working dir: C:\Users\vgami\OneDrive\Desktop\3rd Year\ODD\ASE\ASE-9\.scannerwork
15:37:14.014 INFO Load project settings for component key: 'DuplicateCode'
```

```

15:37:21.326 INFO Start fetching files for the text and secrets analysis
15:37:21.526 INFO Using Git CLI to retrieve dirty files
15:37:21.605 WARN Retrieving only language associated files, make sure to run the analysis inside a git repository to make use of inclusions specified via
"sonar.text.inclusions"
15:37:21.608 INFO Starting the text and secrets analysis
15:37:21.610 INFO 1 source file to be analyzed for the text and secrets analysis
15:37:21.633 INFO 1/1 source file has been analyzed for the text and secrets analysis
15:37:21.633 INFO Sensor TextAndSecretsSensor [text] (done) | time=837ms
15:37:21.639 INFO ----- Run sensors on project
15:37:21.940 INFO Sensor JaCoCo Aggregate XML Report Importer [jacoco]
15:37:21.940 INFO Sensor JaCoCo Aggregate XML Report Importer [jacoco] (done) | time=1ms
15:37:21.941 INFO Sensor IaC Project Sensor [iac]
15:37:21.943 INFO Sensor IaC Project Sensor [iac] (done) | time=2ms
15:37:21.943 INFO Sensor JavaProjectSensor [java]
15:37:21.946 INFO Sensor JavaProjectSensor [java] (done) | time=3ms
15:37:21.947 INFO Sensor Zero Coverage Sensor
15:37:21.958 INFO Sensor Zero Coverage Sensor (done) | time=11ms
15:37:21.959 INFO Sensor Java CPD Block Indexer
15:37:21.982 INFO Sensor Java CPD Block Indexer (done) | time=22ms
15:37:21.982 INFO ----- Gather SCA dependencies on project
15:37:21.993 INFO Dependency analysis skipped
15:37:21.997 INFO SCM Publisher No SCM system was detected. You can use the 'sonar.scm.provider' property to explicitly specify it.
15:37:22.003 INFO CPD Executor Calculating CPD for 1 file
15:37:22.020 INFO CPD Executor CPD calculation finished (done) | time=15ms
15:37:22.273 INFO Analysis report generated in 229ms, dir size=297.1 kB
15:37:22.582 INFO Analysis report compressed in 310ms, zip size=32.8 kB
15:37:22.678 INFO Analysis report uploaded in 96ms
15:37:22.680 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=DuplicateCode
15:37:22.683 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
15:37:22.683 INFO More about the report processing at http://localhost:9000/api/ce/task?id=d267d2dd-b0b4-4083-bc34-72097f85253c
15:37:22.766 INFO Analysis total time: 9.393 s
15:37:22.837 INFO SonarScanner Engine completed successfully
15:37:22.864 INFO EXECUTION SUCCESS
15:37:22.864 INFO Total time: 21.590s

```

Step 9: Open the SonarQube Project Dashboard

Open:

`http://localhost:9000`

Then go to:

Projects → DuplicateCode

Open the project Overview.

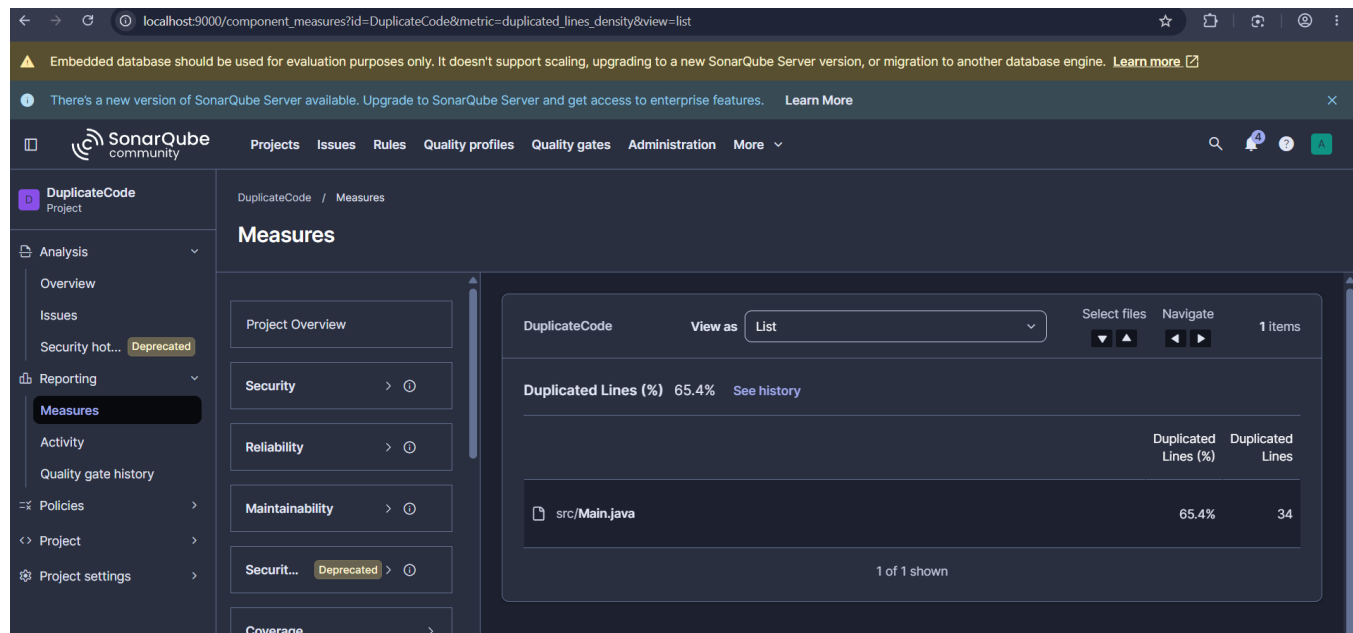
The screenshot shows the SonarQube web interface. At the top, there are navigation tabs: Projects, Issues, Rules, Quality profiles, Quality gates, Administration, and More. The 'Projects' tab is active. On the left sidebar, there are filters for Quality gate (Passed: 2, Failed: 0) and Security (0 info, 0 low, 0 medium, 0 high, 0 blocker issues). The main content area displays two project cards. The first card is for 'Banking' (Public), which has been analyzed 32 minutes ago and passed. It shows a quality score of 0 for Security, 1 for Reliability, and 9 for Maintainability. The second card is for 'DuplicateCode' (Public), analyzed 2 minutes ago and also passed. It shows a quality score of 0 for Security, 0 for Reliability, and 20 for Maintainability. Both cards show 0.0% coverage and 0.0% duplications.

Step 10: Check Duplicate Code

In the dashboard, check the Duplications metric.

For this experiment, SonarQube should identify the repeated code and show a non-zero Duplications percentage.

The exact percentage depends on the code and SonarQube's CPD analysis. Record the percentage shown on your own dashboard.



The screenshot shows the SonarQube web interface for a project named 'DuplicateCode'. The 'Measures' section is selected in the left sidebar. The main content area displays the 'DuplicateCode' project overview, including a 'View as' dropdown set to 'List', 'Select files' and 'Navigate' buttons, and a table showing the 'Duplicated Lines (%)' metric. The table lists one item: 'src/Main.java' with a duplicated lines percentage of 65.4% and 34 duplicated lines. The interface also includes a top navigation bar with links to Projects, Issues, Rules, Quality profiles, Quality gates, Administration, and More. A sidebar on the left contains links to Analysis, Reporting, Measures, Activity, and Quality gate history. A bottom sidebar contains links to Policies, Project, and Project settings.

File	Duplicated Lines (%)	Duplicated Lines
src/Main.java	65.4%	34

What is CPD?

CPD means Copy-Paste Detection. It is the SonarQube analysis used to find sufficiently large repeated code blocks. The Duplications metric shows the amount of duplicated code detected.

Observation

The Java program containing repeated code was analyzed using SonarQube. SonarQube's Copy-Paste Detection identified the repeated code, and the Duplications metric was displayed in the project dashboard.

Result

Thus, duplicate code was successfully analyzed and detected using SonarQube.